
Quiz 1- Section 1

Important Notes: By submitting this exam, you affirm that you have not given or received any unauthorized assistance during this test, and that all work is your own. Academic integrity is highly valued in this course, and I am committed to ensuring a fair testing environment for everyone. Cheating undermines trust and devalues the hard work of others. Be advised that any form of cheating will not be tolerated and will result in disciplinary action, up to and including failure of the course.

Exam Guidelines

- This exam is closed book and notes and **the use of the internet is strictly prohibited!** However, you are free to use a hand-written, double-sided, A4 size “cheat” sheet with your notes.
- **Communication among students is not allowed** during the exam. This includes talking, text messaging, emailing, chatting, forum discussions, etc. Headphones are also not permitted.
- All questions, irrespective of their nature, should be directed only to Prof. Kaya or the teaching assistants.

Submission Guidelines and Deadlines

- When the exam concludes, all papers and the handwritten cheat sheet must be submitted to the TAs or Dr. Kaya.
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A company manufactures two types of cake mixes, A and B , using two raw materials, R_1 and R_2 . The following table summarizes the production requirements, resource availability, profits, and demand limits.

Raw material	Units needed to make 1 unit of		Units available
	A	B	
R_1	1	2	6000
R_2	2	1	8000
Net profit per unit	\$7	\$5	
Maximum demand	3500	2500	

- [20 points] Formulate a linear programming model to determine how many units of A and B the company should produce in order to maximize total profit. Clearly define the decision variables, objective function, and all constraints.
- [20 points] Graph the feasible region of the problem in the two-dimensional plane. Clearly label all constraints and indicate the feasible set/region. Solve the problem graphically and determine:
 - the optimal production quantities of A and B ,
 - the maximum achievable profit.
- [15 points] At the optimal solution, identify which constraints are binding and which are non-binding. Briefly explain what this means in practical terms (in real life).
- [25 points] Find the dual of this problem. Interpret the shadow price associated with each raw material constraint, R_1 and R_2 . Explain what each value represents economically.
- [20 points] Suppose this company has a set of cake mixes C and a set of raw materials R . Each cake mix $c \in C$ yields net profit of r_c , and requires $\alpha_{r,c}$ units of each raw material $r \in R$. Furthermore, each resource r available in capacity β_r . Write the general linear program formulation in closed algebraic form using sets, indices, and parameters.

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Solution

(a) Linear Programming Formulation (20 points)

Decision variables

x_A = number of units of cake mix A produced,
 x_B = number of units of cake mix B produced.

Objective function

$$\max Z = 7x_A + 5x_B$$

Constraints

Raw material availability:

$$\begin{aligned} x_A + 2x_B &\leq 6000 \quad (R_1), \\ 2x_A + x_B &\leq 8000 \quad (R_2). \end{aligned}$$

Demand limits:

$$\begin{aligned} x_A &\leq 3500, \\ x_B &\leq 2500. \end{aligned}$$

Non-negativity:

$$x_A, x_B \geq 0.$$

- Defining the decision variables in a clear way: 5
 - If they do not write down that it is the **number** of cake mixes to be produced: -1
- Defining the objective function in a clear way: 5
 - If they do not explain it verbally: -1
 - If there is a mistake in the coefficients: -1 (be careful this mistake may carry for the rest of the solution!)
- Raw material constraints: 6
 - If they do not explain the constraint verbally: -0.5 for each constraint
 - If there is a mistake in the coefficients: -0.5 for each constraint (be careful this mistake may carry for the rest of the solution!)
- Demand and non-negativity constraints: 4
 - If they do not explain the constraint verbally: -0.5 for each constraint except the non negativity
 - If there is a mistake in the coefficients: -0.5 for each constraint (be careful this mistake may carry for the rest of the solution!)

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(b) Graphical Solution (20 points)

The feasible region is defined by the following constraints and in two-dimensional plane should look like the graph below:

$$x_A + 2x_B \leq 6000,$$

$$2x_A + x_B \leq 8000,$$

$$x_A \leq 3500,$$

$$x_B \leq 2500,$$

$$x_A, x_B \geq 0.$$



The corner points of the feasible region are evaluated below:

Point	x_A	x_B	$Z = 7x_A + 5x_B$
(0,0)	0	0	0
(0,2500)	0	2500	12,500
(1000,2500)	0	2500	19,500
(3500,0)	3500	0	24,500
(3500,1000)	3500	1000	29,500
(3333.33, 1333.33)	3333.33	1333.33	30,000

The optimal intersection (corner) point is found by solving:

$$x_A + 2x_B = 6000,$$

$$2x_A + x_B = 8000.$$

Optimal solution:

$$x_A^* = 3333.33, \quad x_B^* = 1333.33,$$

$$Z^* = \$30,000.$$

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- Correct graph and labeled constraints: 8
 - If there is a major problem with the graph but they have attempted solving: -6 (be careful this mistake may carry for the rest of the solution!)
 - If they did not label the constraints: -1
- Feasible region identified: 2
- Corner point evaluation: 6
 - If they could not correctly identify all 6 corner points, but did 4 of them: -2
- Optimal solution and profit: 4

(c) Binding and Non-binding Constraints (15 points)

At the optimal solution:

- **Binding constraints:**

$$x_A + 2x_B = 6000, \quad 2x_A + x_B = 8000.$$

- **Non-binding constraints:**

$$x_A \leq 3500, \quad x_B \leq 2500.$$

Interpretation: Both raw materials are fully utilized at the optimum, while demand limits do not restrict production. Additional raw materials would allow higher profits.

- Binding constraints identified: 7
- Non-binding constraints identified: 4
- Interpretation: 4

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(d) Dual Problem and Shadow Prices (25 points)

Let:

y_1 = shadow price of raw material R_1 ,

y_2 = shadow price of raw material R_2 .

Dual formulation

$$\min 6000y_1 + 8000y_2$$

subject to

$$y_1 + 2y_2 \geq 7 \quad (\text{cake mix A}),$$

$$2y_1 + y_2 \geq 5 \quad (\text{cake mix B}),$$

$$y_1, y_2 \geq 0.$$

Since both primal constraints are binding, the corresponding dual constraints are tight:

$$y_1 + 2y_2 = 7,$$

$$2y_1 + y_2 = 5.$$

Solving yields:

$$y_1 = 1, \quad y_2 = 3.$$

Interpretation:

- One additional unit of raw material R_1 increases maximum profit by \$1.
- One additional unit of raw material R_2 increases maximum profit by \$3.
- Correct dual formulation: 10
 - Minor coefficient mistake: -1
 - Major mistake: - 5
- Correct shadow prices: 8
 - Could not use complementary slackness to make constraints tight: -4
- Clear economic interpretation: 7
 - Included an explanation, but it's not clear: -2
 - The interpretation is completely wrong: -5

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(e) General Linear Program Formulation (20 points)

Sets

C = set of cake mixes,

R = set of raw materials.

Parameters

r_c = profit per unit of cake mix $c \in C$,

$a_{r,c}$ = units of raw material r required for cake mix c ,

β_r = available capacity of raw material $r \in R$.

Decision variables

x_c = units of cake mix c produced.

General LP

$$\max \sum_{c \in C} r_c x_c$$

subject to

$$\sum_{c \in C} a_{r,c} x_c \leq \beta_r \quad \forall r \in R,$$

$$x_c \geq 0 \quad \forall c \in C.$$

- Sets, parameters, and indices are defined: 6
 - If they introduced a demand parameter, that's great!
 - If they did not introduce a demand parameter, that's also okay.
- Objective function: 6
- Constraints: 6
- Clarity and notation: 2